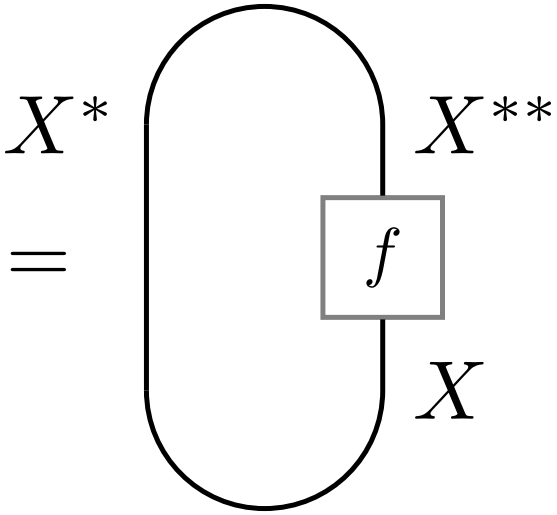$$\mathrm{tr}^R(f) =$$